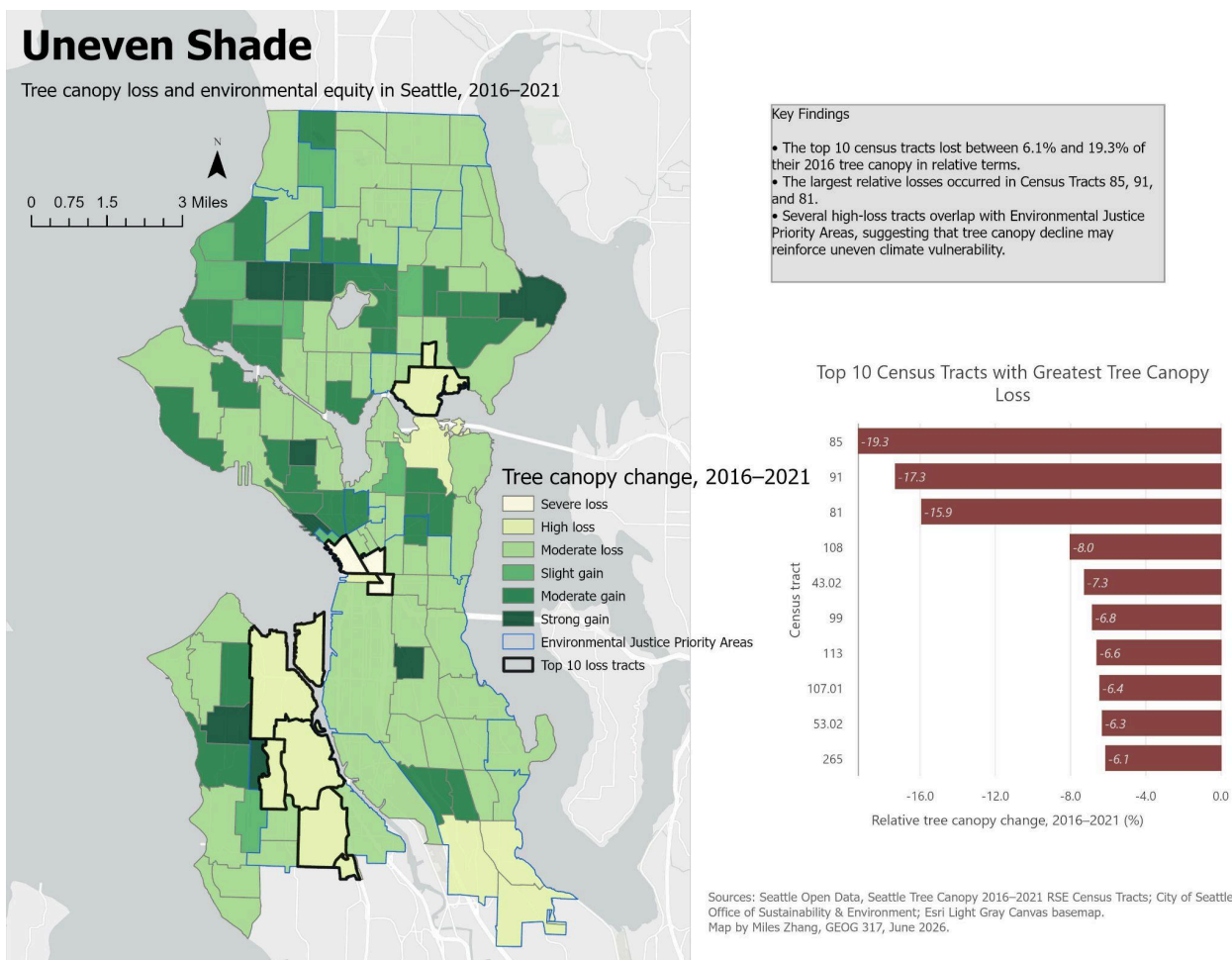




Report

Uneven Shade: Tree canopy loss and environmental equity in Seattle, 2016-2021

Urban tree canopy is not only a landscape feature; it is civic infrastructure. Trees moderate urban heat, filter air pollution, store carbon, absorb stormwater, and make streets more livable. In Seattle, these benefits are also an environmental equity issue because shade is not distributed evenly across neighborhoods. The City of Seattle's 2021 Tree Canopy Assessment reported that citywide canopy cover declined from 28.6% in 2016 to 28.1% in 2021, a net loss of 255 acres (City of Seattle Office of Sustainability & Environment, 2023). This project asks where Seattle experienced the greatest tree canopy decline between 2016 and 2021 and whether those losses overlap with Environmental Justice Priority Areas.

The visual investigation uses the Seattle Tree Canopy 2016-2021 RSE Census Tracts dataset from Seattle Open Data. This dataset provides census-tract-level information on existing canopy, possible canopy, relative percentage change, and racial and social equity indicators

(City of Seattle Open Data, 2025). I mapped relative tree canopy change by census tract, overlaid Environmental Justice Priority Areas in blue, and highlighted the ten census tracts with the greatest relative canopy loss using black outlines. The accompanying bar chart ranks these ten tracts by relative canopy change, allowing the reader to compare the magnitude of loss across locations rather than relying only on the map.

The results show that canopy loss is spatially uneven. The ten highest-loss census tracts lost between 6.1% and 19.3% of their 2016 tree canopy in relative terms, with the largest losses occurring in Census Tracts 85, 91, and 81. These values are relative percent changes, not percentage-point changes. For example, a tract with a -19.3% relative change did not necessarily lose 19.3 percentage points of land-cover canopy; rather, its 2021 canopy was 19.3% lower than its 2016 canopy baseline. This distinction matters because relative change highlights where local canopy systems changed most sharply, even when the citywide decline appears modest.

The equity overlay adds the central policy insight. Several high-loss tracts overlap with Environmental Justice Priority Areas, suggesting that canopy decline may compound existing patterns of vulnerability. Lower-canopy areas can experience greater exposure to heat and pollution, while residents may have fewer resources to adapt or advocate for tree protection. American Forests' Tree Equity Score framework uses a similar logic by evaluating tree canopy alongside factors such as surface temperature, income, employment, race, age, language, and health (American Forests, n.d.). In this sense, the map treats tree canopy as urban infrastructure connected to climate resilience and public health, not simply as neighborhood greenery.

This spatial analysis is useful for Seattle urban forestry planners, environmental justice organizations, neighborhood advocates, and policymakers deciding where to target limited resources. Rather than treating Seattle's canopy goal as only a citywide percentage, the map identifies where protection, planting, outreach, and maintenance should be prioritized. Seattle's goal of reaching 30% canopy cover by 2037 should therefore be evaluated not only by total acres gained, but also by whether gains occur in communities with the greatest need (City of Seattle, n.d.).

References

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